



**SUNY POLYTECHNIC
INSTITUTE**

Rocketry Detailed Design Report

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Overall Rocket Design

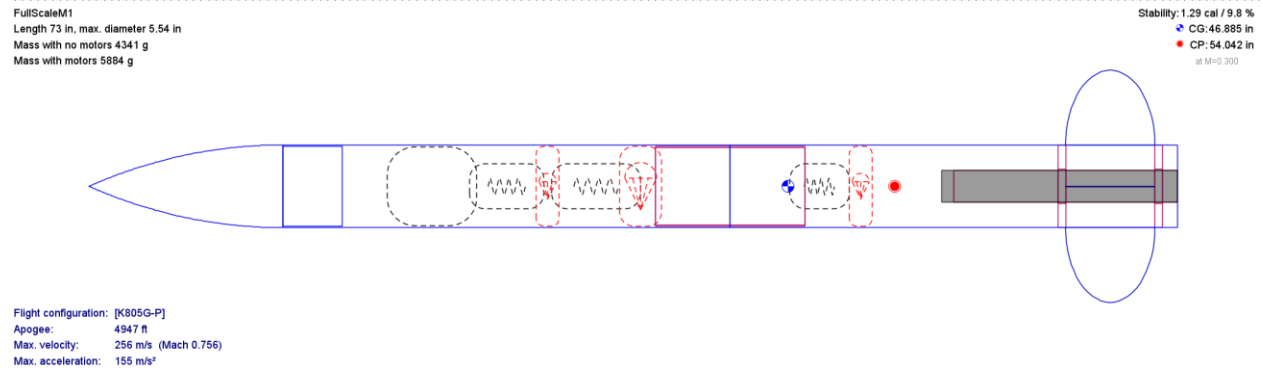


Figure 1: Openrocket outline of full-scale general layout

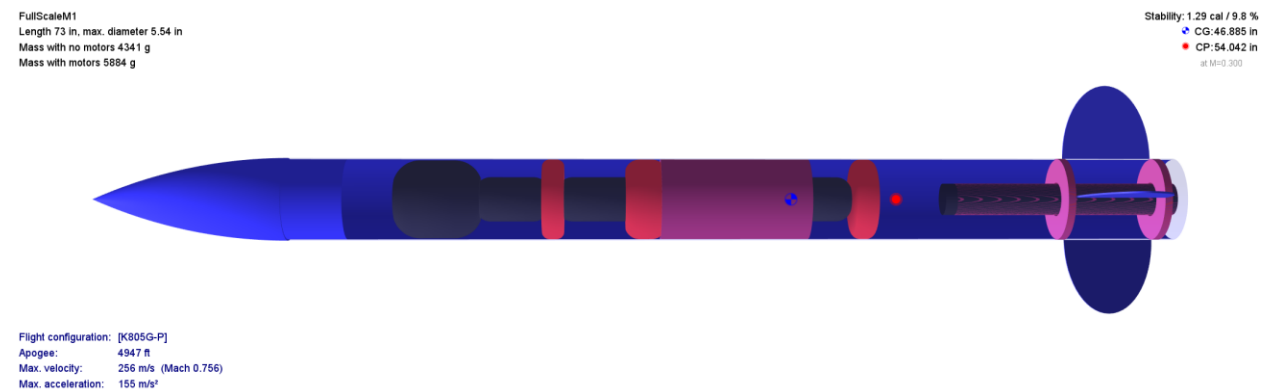


Figure 2: 3D view of full-scale model

Current layout for the full-scale rocket with each of its systems attached: airframe, payload, and avionics

Body material:



Figure 3: Blue Tube of various diameters

For the full-scale rocket, the team will be utilizing a 5.5” inner diameter 72” length Blue Tube as the material of choice for the airframe. While consideration was given to another common airframe material – fiberglass - the team opted to not utilize it as it would require different tools than are available on site, is more fragile, and poses more of a health risk due to the particulates it creates.

Blue Tube allows the team to fabricate the airframe of the rocket with the tools at our disposal in the CGAM labs, along with not posing much of a health risk as the team will be wearing the proper PPE when fabricating.

Should the rocket sustain damage to the airframe, Blue Tube creates dents at the site of damage which allows the team to determine the full extent of forces experienced, and if the part can be reusable as a result.

Nose Cone:



Figure 4: LOC Nose cone

For the nose cone, the team decided to purchase a 5.32" diameter polyethylene plastic nosecone as it fits neatly into the airframe.

The option to fabricate a custom nose was acknowledged, however it was decided against given the lack of experience with nose cone fabrication and lack of adequate equipment for such fabrication.

The nose cone will not be fixed to body tube as it has a tight friction fit to not be removed during flight except being acted upon by the ejection charges. As there will be objects in the way of the nosecone and charges, having less connections allow for easier deployment and testing.

Coupler:



Figure 5: Coupler from sub-scale launch

The coupler portion of the rocket is to be fabricated with a 5.5" full length coupler Blue Tube. The coupler allows for the connection of the lower half with the upper half of the airframe during launching of the rocket and allows for the anchoring points in the form of eyebolts for the soft

link connection upon separation after apogee. The bulkheads for the coupler are fabricated out of particle board as its lightweight yet sufficient for handling the force load to be experienced.

The coupler also houses the avionics bay where the altimeters and GPS are fastened to a particle wood bed. The wood bed is then bolted together with threaded rods.



Figure 6 – Shear pins

Shear pins are to be used for connecting the lower and upper halves of the body during the first half of the flight and separating during the drogue parachute's ejection via the black powder as they are designed to shear from the forces exerted by the powder.

The figure above demonstrates the sub-scale coupler with the eyebolts on both ends along with the electrical wires sticking out from where the black powder charges were connected. The subscale material for the body of the coupler was comprised of cardboard, thus the visual difference from the Blue Tube.

Anti-Zippering Donut:



Figure 7: Anti-Zippering Donut

During parachute deployment, one concern is the friction between shock cord and body tube friction causing the body tube to break and zipper. This anti-zipper donut was made to keep the shock cord centered away from the body tube at a specified length. The design allows for multiple holes to be used for tying and ease of modification in modeling software through the center hole dependent on thickness of shock cord.

Avionics Bay:

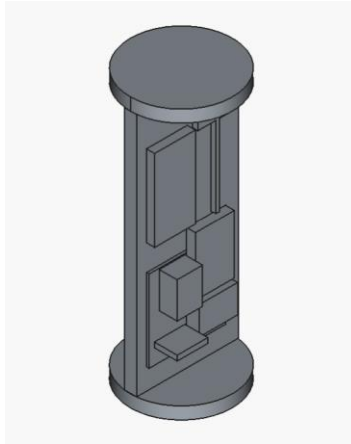


Figure 8: Layout of final avionics

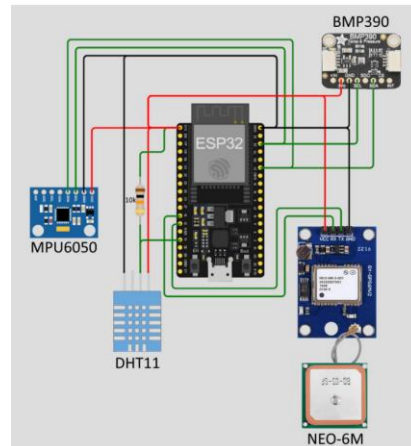


Figure 9: ECE electronics connected

The avionics bay is inside the coupler with a hole drilled into the side of the tube to allow for sensors to detect changes in atmosphere. Along with the Blue Raven altimeter, a gyroscopic, temperature, humidity, and barometric pressure sensors with a GPS module connected to an ESP32 board have been added. These sensors have been added to allow for real time feedback and prediction during flights, which the Blue Raven Altimeter does not provide as its data has to be downloaded after. LoRa is being used on the boards to be able to conduct long distance communication of 360 miles with a transit time of 63.195ms.

Fin Design:

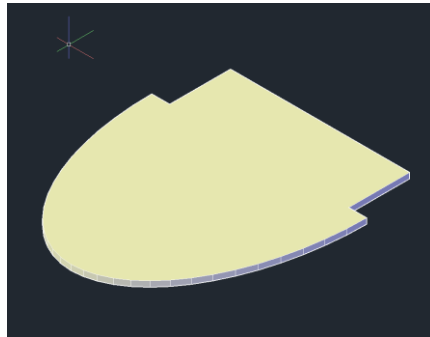


Figure 12: Fin Design

- Number of fins: 4
- Root Chord: 6"
- Fin Height: 5.25"
- Fin Tab Length: 5"
- Fin Tab Height: 1.7"

A recommended stability score for a successful model rocket launch should be between 1-2. Using 4 fins allowed for the maximum amount of stability to be imparted, whereas a 3-fin design yielded a stability score of 0.6 which is well below the recommended range.

The fin tab behind the fin is to allow the fins to be slotted into the rocket for ease of assembly. 1.7" is the distance between the outer diameter of the body tube and motor mount tube to fully utilize area coverage. 0.5" were taken of each side of the tab so that they can be slotted between the centering rings of the motor mount tube to ensure that each fin and the motor mount are level with the rocket.

The fins are planned to be made of fiber glass infused with layers of carbon fiber and is to be fabricated in the 3D printing laboratory.

Parachutes:



Figure 13: 24-in Drogue Parachute



Figure 14: 48-in Main Parachute

The team decided to purchase a set of parachutes for drogue and main: 24-in and 48-in for each respectively.

Main parachute is hand calculated to experience a drag force of 69 lbs. with the simulation yielding an expected drag force of 78 lbs. The discrepancy between the two calculations most likely derives from slightly different coefficient of drag being utilized in the simulation and what was used empirically on hand.

Additionally, the deviation in drag force can be attributed to the simulation taking into consideration the entirety of the rocket whereas the hand calculations solely focused on the parachutes with forces such as weight being simplified into a singular point.

Equation for parachute force – Main Parachute

$$\text{Drag Force Equation: } F_d = \frac{1}{2} \rho C_d A V^2$$

$$\text{Area: } A = \pi r^2 = 1809.6 \text{ in}^2 = 1.167 \text{ m}^2$$

$$\text{Density of air: } \rho = 1.225 \text{ kg/m}^3$$

$$\text{Coefficient of Drag: } C_d = 2.2$$

$$\text{Velocity Upon Decent: } V = -14 \text{ m/s}$$

The velocity experienced by the main parachute is estimated to be -14 m/s as the main parachute is expected to deploy at 1,000 ft. The coefficient of drag was given by manufacturer's website.

$$F_d = \frac{1}{2} \times 1.225 \frac{\text{kg}}{\text{m}^3} \times 2.2 \times 1.167 \text{ m}^2 \times \left(14 \frac{\text{m}}{\text{s}}\right)^2$$

$$F_d = 308 \text{ N} = 69.2 \text{ lbs.}$$

Due to the drogue parachute is deployed at flight apogee, the velocity experienced by the parachute is zero so there is no opening shock force. As the rocket descends, the velocity increases to the maximum of -14 m/s before the main parachute deploys and further slows down the rocket to 5.42 m/s.

The 48-in parachute is expected to handle a maximum of 12.5 pounds at a descent rate velocity of 20 ft/s, with the 24-in parachute being capable of handling 2.2 pounds at 20 ft/s. The gross weight of the rocket on the pad is 14.4 lbs; however, this weight drops down to 10.3 lbs. at apogee due to the motor fuel having been spent.

Separation method:



Figure 15: Spent black powder charge from testing

Black powder charges are being used to eject the drogue and main parachute at a given height. These were chosen over CO₂ cartridges due to its ease of implementation, compatibility with Blue Raven Altimeters, and lower weight. Utilization of CO₂ adds weight and size constraints,

along with adding more complexity to the activation of the cartridges in the air. Black powder is able to achieve the same forces required for the successful shearing of the shear pins while not adding much in weight, complexity, and cost.

The following formulas were used to determine the necessary amount of black powder needed to successfully break the shear pins connecting the rocket to release the parachutes.

Formula 1: Pressure

$$\text{Pressure (psi)} = \frac{\text{Force (lbs)}}{\text{Area (Inch}^2\text{)}}$$

Simulation given pressure: 14.46psi

Formula 2: Volume of Airframe – Diameter is inside diameter and length is the unified length

$$\text{Volume (inches}^3\text{)} = \frac{\pi \times (\text{diameter(inches)})^2 \times \text{Length(inches)}}{4}$$

Diameter = 5.38 in

Volume = 682.75 in³

Length = 30 in

Formula 3: Black powder required for each section

$$\text{Grams (BP)} = \frac{(454 \text{ grams})}{1 \text{ lbf}} \times \frac{\text{Pressure(psi)} \times \text{Volume (inches}^3\text{)}}{266 \frac{\text{inches} \times \text{lbf}}{\text{lbfm}} \times 3307 \text{ }^\circ\text{R}}$$

Grams (BP) = 5.09 g

Data supported during sub scale flight for 3” diameter was used and 1.6g calculated and successfully deployed during sub-scale flight. It is important to note that these calculations are meant to be used as a start for ground testing and not on the day of launch. The amount of black powder needed may change due to the amount of shear pins being used and if there are obstructions inside the rocket that receive the forces instead. There are 2 sets of charges. One set is used as main charges to deploy at given values and the second set as backup on a timer for 5 seconds after main charges in case they did not have the necessary strength to fully deploy.

Payload Design:

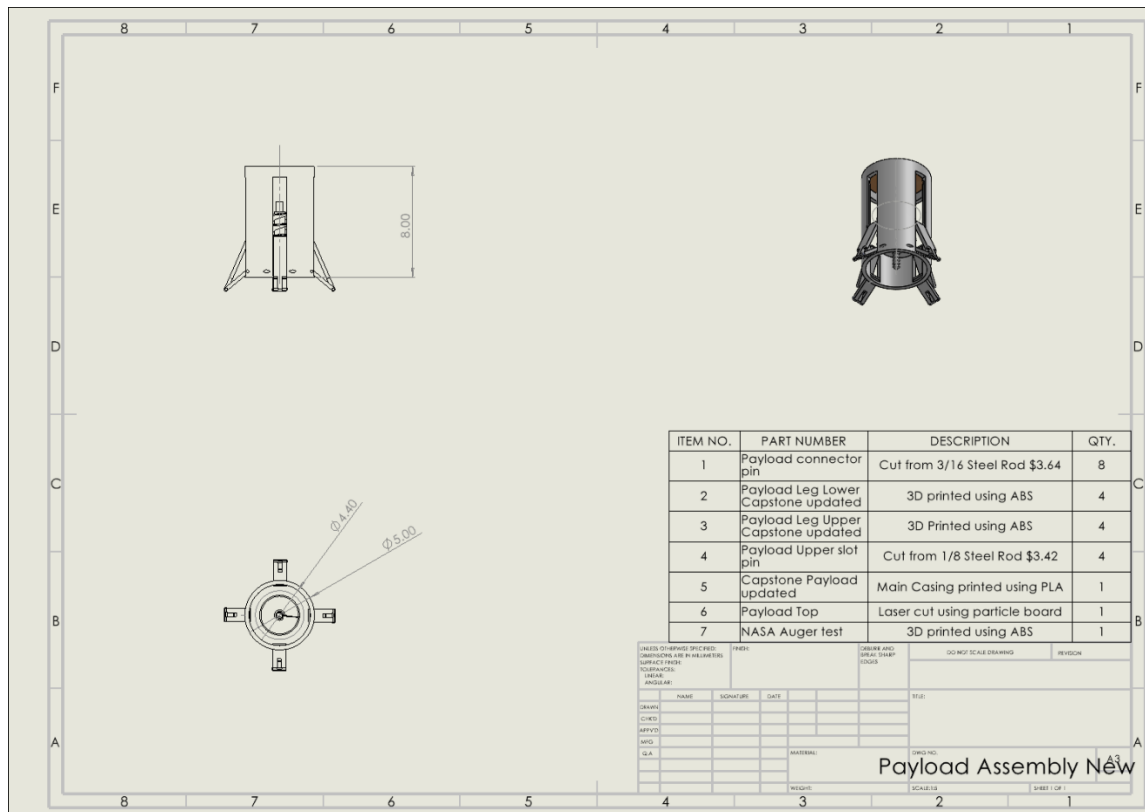


Figure 16 Payload Section View

Payload Casing:

The payload casing diameter is 5.1” This is to give some clearance for the payload legs to slightly stick out inside the rocket. The casing itself is 8” tall, giving adequate space for the auger and electronics stored within. Within the casing there are 1/8” slots on both sides of each opening so the legs can slide to an open position. The casing will be made of PLA.

Payload Legs:

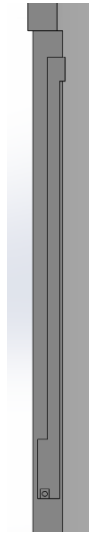


Figure 17: isolated view of payload sliding slots

The legs are comprised of two separate parts, an upper and lower leg. Together they are connected via a 3/16" rod. The legs are to go to a deployed position once the payload leaves the rocket. To accomplish this 1/8" tension springs within the slots (Figure 17) on the side will pull the upper leg down to force the mechanism down. The legs will be made using TPU, since it is more shock absorbent, as compared to PLA.

Auger

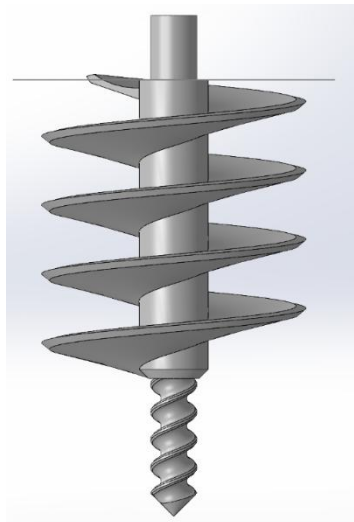


Figure 18 Auger Side View

The auger (Figure 18) is 3.6" in diameter, 5.45" tall. The tip of the auger is 2.2". The auger is to be 3D printed using ABS as well. The auger's tip is smaller to initially break into the dirt, then the larger diameter part is to collect dirt to begin the soil analysis. The auger is to be powered by a motor provided by the MET team.

Payload Top:

The payload top is made of 4.65" diameter particle board; it is to be JB welded top of the casing. Particle board is used since we will have to drill into it to attach the 3/16" eyehook for the parachute.

pH Sensor:



Figure 19: RS485 Soil Sensor

The RS485 soil sensor (Figure 19) was chosen by the ECE team, as it's something they can modify to work on our project, and the sensor is small and can easily fit within the entire payload. Figure 7 is a diagram documenting how the ECE's plan to gather inputs and turn them into readable outputs.

ECE Payload:

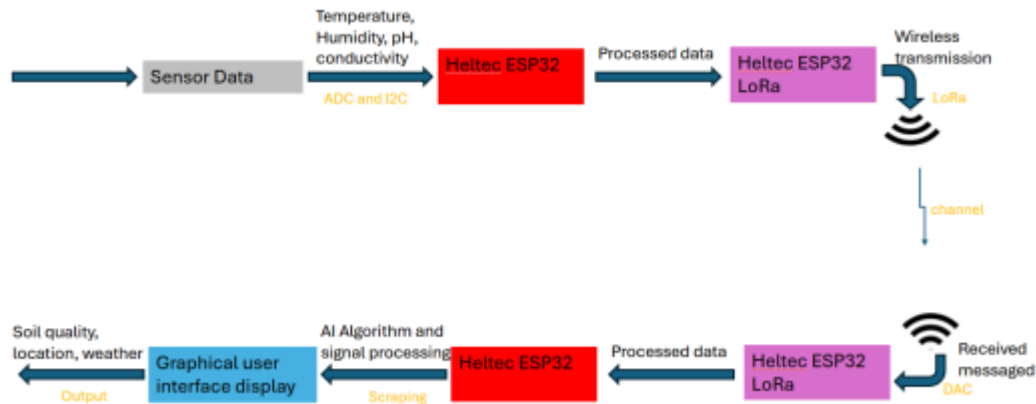


Figure 20: ECE setup

Figure 20 provides a visual guide of what the ECE team aims to do for their capstone specific payload. Their capstone requires them to wirelessly transmit the data collected from the rocket, process the data and to utilize AI in some capacity, and to output it in the form of a graphical interface display. The greater scope of their own project requirements are outside the scope of this report.

Bill of Materials

Item	Link	Quantity	Price
5.5" D Blue Tube 72"	https://alwaysreadyrocketry.com/product/1-15-29mm-x-062-wall-x-48-airframe-mmt/	1	\$127.30
5.5" full length coupler 48"	https://alwaysreadyrocketry.com/product/1-15-29mm-x-062-wall-x-48-coupler/	1	\$61.62
2.15" (54mm) x 24" MMT	https://alwaysreadyrocketry.com/product/1-15-29mm-x-062-wall-x-48-airframe-mmt/	1	\$13.95
5.38" Nosecone Standard	https://locprecision.com/collections/rocket-components/products/nose-cones?variant=39778657239231	1	\$64.83
54/1706 motor casing complete set *used*	https://performancehobbies.com/secure/store.aspx?groupid=1343	1	\$206.39
2800lb 20ft Shock Cord	https://the-rocketman.com/braided-kevlar-shock-cord/	1	\$41.00
24" elliptical parachute	https://shop.fruitychutes.com/collections/classic-elliptical-12-to-60/products/24-elliptical-parachute-2-2lb-20fps	1	\$91.00

48" elliptical parachute	https://shop.fruitychutes.com/collections/iris-ultra-standard-chutes/products/iris-ultra-48-standard-parachute-12-5lbs-20fps	1	\$161.21
1515 Rail Button	https://www.apogeerockets.com/Building_Supplies/Launch_Lugs_Rail_Buttons/Rail_Buttons/1-5_Rail_Button_Large	1	\$7.49
K805G-P Motor	https://www.apogeerockets.com/Rocket-Motors/AeroTech-Motors/54mm-Propellant-Kits/54-1706-Kits/Aerotech-54mm-Propellant-Kit-K805G-PS	1	\$179.96
18x18 Fire Blanket	https://locprecision.com/products/locprecision-flame-resistant-blankets?variant=39778640429247	1	\$13.58
Jb Weld 25mL 2 Pack	https://www.amazon.com/gp/product/B0DRW7VX6B/ref=ewc_pr_img_2?smid=ATVPDKIKX0DER&th=1	1	\$13.29
Shear pins 20 pack	https://www.apogeerockets.com/Building-Supplies/Misc-Hardware/Nylon-Shear-Pins-20-pack	1	\$4.63
Soft Link	https://www.apogeerockets.com/Building-Supplies/Recovery-Equipment/Quick-Links/Small-Soft-Link	1	\$14.29
Blue Raven Altimeter	https://www.featherweightaltimeters.com/store/p25/Blue_Raven_Permalink.html	2	\$175.00
E-match 20 pack	https://www.csrocketry.com/recovery-supplies/ejection-supplies/firewire-electric-match.html?srsId=AfmBOopUGawbLiV84kA5QeDv3OPN5pASwX1mG3_itw2CvJcVTBbbqBNO	1	\$16.00
GPS Tracker Set	https://www.featherweightaltimeters.com/store/p22/Featherweight_GPS_Tracker_%28upd%29.html	1	\$371.00
RS485 Soil Sensor	https://www.amazon.com/Output-Temperature-Humidity-Moisture-Sensor/dp/B0BVW1KNM4	1	\$39.55

Total	\$1777.20
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